

Assisted Living Monitor

User Guide - Version 1.0 - March 2026

What It Does

The Assisted Living Monitor detects unexpected inactivity in the home. It learns the resident's normal daily patterns over 7 days, then raises an alert when no movement is detected during a time the system predicts someone should be present.

You interact with the system in two ways:

- The Monitor web UI - view status, adjust settings, manage Vacation Mode
- Node-RED - receive and act on MQTT alerts

The Web UI

Detection Settings

The Settings panel displays the active detection parameters:

Setting	Default	Meaning
Forecast threshold	0.70	Min AI confidence that presence is expected
Silence threshold	30 min	Inactivity duration before an anomaly fires
Emergency threshold	2 h	Extended absence that escalates to EMERGENCY
Alert throttle	60 min	Minimum gap between repeated alerts per room
MQTT topic	aragon/assistedl	Topic where alerts are published

Event History

The Events panel shows a timestamped log of all anomaly alerts, including room, severity, and the duration of silence at the time the alert was raised.

Vacation Mode

Enable Vacation Mode whenever the resident is away from home to prevent false alerts.

To enable:

1. Open the Assisted Living section in the web UI
2. Toggle Vacation Mode ON
3. Optionally enter a reason (e.g. "Holiday - returns 20 March")

To disable:

- Toggle Vacation Mode OFF when the resident returns home

NOTE: If Vacation Mode is NOT enabled and the home is empty, EMERGENCY alerts will fire after 2 hours of silence.

How Detection Works

Every few minutes, the system checks two conditions per monitored room (Bedroom and Living Room):

- Forecast ≥ 0.70 : the AI model expects someone to be present right now
- Silence ≥ 30 min: no motion detected for at least 30 minutes

When BOTH conditions are met simultaneously, an alert is published to MQTT.

If silence exceeds 2 hours, severity escalates to EMERGENCY regardless of the forecast score. Repeated alerts per room are throttled - at most once every 60 minutes.

MQTT Alerts in Node-RED

Topic

aragon/assistedliving/anomaly

Example Payload

```
{
  "severity": "EMERGENCY",
  "room": "Bedroom",
  "forecast": 1.000,
  "silent_minutes": 121,
  "vacation_mode": false
}
```

Suggested Node-RED Flow

1. MQTT In node - set topic to: aragon/assistedliving/anomaly
2. JSON node to parse the payload
3. Switch node - route by "severity" field:
 - "EMERGENCY" -> push notification, SMS, or email
 - Other -> log to dashboard or home automation flow
4. Optional: filter on vacation_mode == false to suppress holiday alerts

Quick Reference

Situation	Action
Resident going on holiday	Enable Vacation Mode in the UI before they leave
Resident returns home	Disable Vacation Mode in the UI
Too many repeated alerts	Alerts throttled to max once per hour per room
No alerts in Node-RED	Verify MQTT topic: aragon/assistedliving/anomaly
System just installed	Wait 7 days for the learning period to complete